Linear Algebra and Calculus Lecture 3
Continuity and Derivatives

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Continuity

We define the notion of continuity using limits.

Definition (1.4.4)**.** A function $f : D \to \mathbb{R}$ is **continuous** at $c \in D$ if $\lim_{x\to c} f(x) = f(c)$. If the equality is not satisfied or the limit does not exist, we say that f is **discontinuous** at c.

Definition (1.4.7)**.** We say that a function f is **continuous on the interval** I if it is continuous at each point of I. In particular, we will say that f is a **continuous function** if it is continuous at every point of its domain.

Example. Some examples of continuous functions are

- Polynomials
- Rational functions
- Rational powers: $x^{m/n} = \sqrt[n]{x^m}$
- Trigonometric functions
- Absolute value function $|x|$

Example. The function

$$f(x) = \begin{cases} 3x, & x \leq 2 \\ \frac{12}{x}, & x > 2 \end{cases}$$

is continuous at $x = 2$, since $\displaystyle\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} 3x = \lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} \frac{12}{x} = f(2) = 6$.

Theorem (1.4.6)**.** *Let f and g be functions continuous at a point of the domain c, the following are also continuous at c:*

1. *$f + g$ and $f - g$*
2. *$f \cdot g$*
3. *$k \cdot f$, for any $k \in \mathbb{R}$*

4. $\frac{f}{g}$ *if* $g(c) \neq 0$

5. $(f)^{\frac{1}{n}}$*, if* $f(c) > 0$ *for* n *even*

6. The composition $f \circ g$

Example. The function $f(x) = \sin^2(e^x)$ is continuous in all $\mathbb{R}$ since it is a composition of the functions e^x, $\sin x$ and x^2, which are continuous in all $\mathbb{R}$.

Theorem (1.4.9)**.** ***The Intermediate Value Theorem*** *Let* f *be continuous on* $[a, b]$ *and* $s \in [f(a), f(b)]$*, then* $\exists c \in [a, b]$ *such that* $f(c) = s$*.*

Theorem. ***Bolzano's Theorem*** *Let* f *be continuous on* $[a, b]$ *such that* $f(a)f(b) < 0$ *(that is,* $f(a)$ *and* $f(b)$ *have opposite signs), then* $\exists c \in (a, b)$ *such that* $f(c) = 0$*.*

Example. The function $f(x) = x$ has a zero in $[-1, 1]$, since $f(-1) = -1 < 0$ and $f(1) = 1 > 0$. Since f is continuous, by Bolzano's Theorem, $\exists c \in (-1, 1)$ such that $f(c) = 0$.

Derivatives

If we wanted to know how much a function varies between two points (x_1, y_1) and (x_2, y_2) then we would take the ratio of change $\frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$. In the same manner we can express such ratio in terms of x and an increment h (and their respective images). In such scenario, the rate of change would look like $\frac{f(x+h)-f(x)}{(x+h)-x} = \frac{f(x+h)-f(x)}{h}$.

So if we want to know the slope or instantaneous rate of change of a function at a certain point x_0, we would make this increment h infinitesimally small. Such slope will be the slope of the line *tangent* to $f(x)$ at x_0.

Definition (2.2.4)**.** A function f is **differentiable** at a point a of its domain if its domain contains an open interval containing a, and the following limit exists,

$$L = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

If such limit exists we say the derivative of f at a point x is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

Exercise. If $f(x) = x^r$, prove that $f'(x) = rx^{r-1}$.

Example. If $f(x) = |x|$, then

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \frac{x}{|x|},$$

which doesn't exist for $x = 0$.

There are other ways of notating the derivative of a function, and one of them is the differential of a function, also known as the *Leibniz notation*.

$$f'(x) = D_x f(x) = Df(x) = \frac{d}{dx} f(x).$$

Such notation comes from taking the ratio of increments and making the difference in x very small:

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}.$$

We can also express the evaluation of the derivative at a point by using Leibniz notation:

$$f'(a) = \frac{d}{dx} f(x)\Big|_{x=a}.$$

Theorem (2.3.1). *If f is differentiable at x, then f is continuous at x.*

CAREFUL! The reciprocal is not true. For instance, $f(x) = |x|$ is continuous but not differentiable at $x = 0$. Check also the Weierstrass function.

Theorem (2.3.2). *If f and g are differentiable at x then the following are also differentiable and satisfy:*

- $(f + g)'(x) = f'(x) + g'(x)$
- $(f - g)'(x) = f'(x) - g'(x)$
- $(k \cdot f)'(x) = k \cdot f'(x)$ *for* $k \in \mathbb{R}$

Example.

$$\frac{d}{dx}(x^2 + 3x) = 2x + 3.$$

Theorem (2.3.3). *If f and g are differentiable at x then*

$$(f \cdot g)'(x) = f'(x)g(x) + f(x)g'(x).$$

Example. Applying the product rule for the derivative of the product is the same as first multiplying and then differentiating:

$$\begin{aligned}
\frac{d}{dx}\left((x^2+1)(x^3+4)\right) &= \frac{d}{dx}(x^2+1) \cdot (x^3+4) + (x^2+1)\frac{d}{dx}(x^3+4) = 2x(x^3+4) + (x^2+1)(3x^2) \\
&= 5x^4 + 3x^2 + 8x, \\
\frac{d}{dx}\left((x^2+1)(x^3+4)\right) &= \frac{d}{dx}(x^5 + x^3 + 4x^2 + 4) = 5x^4 + 3x^2 + 8x.
\end{aligned}$$

Theorem (2.3.5). *If f and g are differentiable at x then*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}.$$

Example.

$$\frac{d}{dx}\left(\frac{1 - x^2}{1 + x^2}\right) = \frac{(1 + x^2)(-2x) - (1 - x^2)(2x)}{(1 + x^2)^2} = -\frac{4x}{(1 + x^2)^2}.$$

Chain Rule

Theorem (2.4.6)**.** *If $f(u)$ is differentiable at $u = g(x)$, and g is differentiable at x, then $(f \circ g)(x) = f(g(x))$ is also differentiable at x and $(f \circ g)'(x) = f'(g(x))g'(x)$.*

Example. Let $h(x) = \sin x^3$. Here $g(x) = x^3$ and $f(x) = \sin x$ so that $h(x) = (f \circ g)(x) = f(g(x))$. Since both f and g are differentiable, we have that $h'(x) = f'(g(x))g'(x) = \cos(g(x))3x^2 = 3x^2 \cos x^3$.

Example.

$$\frac{d}{dx}(7x-3)^{10} = 10(7x-3)^9(7) = 70(7x-3)^9.$$

We can also compute the derivative of the derivative:

$$f''(x) = \frac{d}{dx}\frac{d}{dx}f(x) = \frac{d^2}{dx^2}f(x) = D_x^2 f(x).$$

And in general, we can just compute as many derivatives as we want as long as they exist

$$f^{(n)}(x) = \frac{d^n}{dx^n}f(x) = D_x^n f(x).$$

Example. If $f(x) = x^5$, then $f'(x) = 5x^4$, $f''(x) = 20x^3$, $f'''(x) = 60x^2$, $f^{(4)}(x) = 120x$, $f^{(5)}(x) = 120$ and $f^{(6)}(x) = 0$.